EXPERIMENT NO : 11

EXPERIMENT TO UNDERSTAND PANDAS LIBRARY USE CASES IN DATA SCIENCE

AIM:

Experimenting to understand pandas library use cases in Data science.

CODE:

```
import numpy as np
import pandas as pd

# Create DataFrame
list=[['1','Smith',50000],[2,'Jones',60000]]

df=pd.DataFrame(list)
df

# Assign column names
df.columns=['EmpId','Name','Salary']
df

# Display information
df.info()

# Read employee.csv
```

```
df=pd.read_csv("/content/employee.csv")

df.head()

# Display last records
df.tail()

# Display information
df.info()

# Display salary column
df.salary

# Check type
type(df.salary)

# Mean
df.salary.mean()

# Median
df.salary.median()

# Mode
df.salary.mode()

# Variance
df.salary.var()
```

```
# Standard deviation
df.salary.std()

# Descriptive statistics
df.describe()

# Descriptive statistics for all columns
df.describe(include='all')

# Display column names
empCol=df.columns
empCol

# Display values
emparray=df.values
emparray

# Create Employee DataFrame
employee_DF=pd.DataFrame(emparray,columns=empCol)
employee_DF
```

OUTPUT:

```
Saving employee.csv to employee.csv
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Empd    2 non-null      int64
 1   Name    2 non-null      object
 2   Salary  2 non-null      int64
dtypes: int64(2), object(1)
memory usage: 180.0+ bytes
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   emp id  7 non-null      int64
 1   name    7 non-null      object
 2   salary  7 non-null      int64
dtypes: int64(2), object(1)
memory usage: 300.0+ bytes
```

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000
5	6	SRI HARSHAVARDHANAN R	3000
6	7	SRI HARSHAVARDHANAN R	6000

RESULT:

Thus,the given data was analyzed successfully and required output was obtained.